MARCUS ESCALONA

PRODUCT & FORWARD DEPLOYED ENGINEER · FULL-STACK, AI/LLM & VOICE AI
Mexico City, MX · Open to hybrid (CDMX) or remote (USD) · Native English & Spanish
+52 735 106 0241 · marcuspicazo@outlook.com · marcusescalona.art
github.com/MarcusPicazo · linkedin.com/in/marcus-escalona

PROFESSIONAL SUMMARY

Product engineer with 6+ years across product, systems and infrastructure engineering who has taken several products from zero to production. I own problems end to end: scoping with non-technical stakeholders, designing the architecture, building in React, TypeScript, Node.js and Python, integrating third-party APIs and shipping. Recently I built an AI voice and messaging product on LLMs, speech-to-text and telephony/WhatsApp APIs, and a real-time 3D sales platform for real-estate developers. I lead engineering for a 200+ user platform suite and work AI-native, directing agentic tools such as Claude Code to ship fast without trading off quality. Native English and Spanish speaker with a background in physics and audio engineering.

TECHNICAL SKILLS

Frontend & Core React · TypeScript · JavaScript (ES6+) · Zustand / Redux · Tailwind CSS · Vite · Next.js · HTML5 · CSS3
AI / LLM & Voice Claude API (Haiku / Sonnet) · LLM prompting & evaluation · Vapi voice agents · Speech-to-text (Deepgram / Whisper) · OCR (Tesseract.js)
Integrations & APIs Twilio · WhatsApp Cloud API · Resend · Webhooks (signature verification, idempotency) · REST APIs
Cloud & Backend Node.js · Python microservices · Supabase (PostgreSQL, Auth, RLS, Realtime, Edge Functions, pg_cron) · Firebase · Prisma · SQL · CI/CD
3D & Performance Three.js · React Three Fiber · WebGL · Canvas API · Procedural geometry · Custom physics · Render-loop & memory profiling
Tools & Workflow Git · GitHub · Vercel · Netlify · Claude Code · Cursor · AI-native development · Automated testing · Agile

SELECTED PROJECTS (INDEPENDENT)

Socio · Bilingual AI receptionist for US Latino service businesses github.com/MarcusPicazo/SPEC-Socio

- Architected and built the MVP of a bilingual (EN/ES) AI receptionist for Latino-owned contractors in the US: a Vapi voice agent on Twilio numbers answers English calls, and Claude Haiku summarises and classifies each call and sends the owner a Spanish summary over the WhatsApp Cloud API.
- Built the voice-note-to-quote flow: Spanish voice notes are transcribed, turned into itemised English quotes by Claude Sonnet, rendered to PDF in Node.js (React-PDF on Vercel) and delivered by email (Resend) with a public accept link.
- Multi-tenant backend on Supabase (PostgreSQL, RLS, Edge Functions, pg_cron) with verified webhooks, atomic claim-before-process handling against duplicate events and background processing; 111 automated tests passing.
- Designed the customer onboarding (15-minute setup with call forwarding per carrier and a guided test call) and defined pricing and the ideal customer profile.

Tezontle 3D · Real-time 3D sales platform for real-estate developers github.com/MarcusPicazo/aura-demo

- Built a reusable engine for interactive 3D unit selectors for pre-sale developments (React Three Fiber, Three.js, Zustand): towers are generated procedurally from JSON polygon configs, so no 3D modelling is needed, and units are colour-coded by live availability.
- Connected inventory to Supabase Realtime with RLS: when a sales agent marks a unit as sold in the admin panel, it updates on every screen instantly. Added lead capture, WhatsApp deep links per unit, filters, facade view and SVG floor plans.
- Built a sales portal and lightweight CRM (admin and agent roles, inventory, reservations, prospects, audit history) plus a scroll-driven cinematic landing. An engine-first architecture keeps client data in config only, cutting a new client from ~6 weeks to 1–2.
- Built the client-facing demo used to pitch developers, and the packaging of what each client gets.
- Repos: github.com/MarcusPicazo/aura-demo · github.com/MarcusPicazo/aura-landing · github.com/MarcusPicazo/tezontle-web

Bodega Abierta · open-source WMS for food banks (RapiJazz A.C.)
Multi-tenant, offline-first PWA warehouse management system for Mexican food banks, Apache-2.0. Next.js, TypeScript, PostgreSQL and Prisma; donation intake with OCR and barcode scanning (Tesseract.js, jsQR), FEFO expiry alerts, distribution tracking, a donor impact portal and sustainability reporting (kg rescued, waste %, CO_2e). Wrote the product spec and UX for ~35 screens; in development.

NIMIO Levels github.com/MarcusPicazo
Gamified team progression platform with code-based authentication, role-scoped access and an audited action history. React, TypeScript and Supabase (PostgreSQL) — access codes stored hashed rather than in plaintext, with row-level security enforcing per-member scope.

Taylor · Safe mobility for women (co-founder) apptaylor.com
Core architecture for pilot programmes in Mexico City, California and London, integrating geolocation logic and real-time push notifications. B2B2C model with a live Android app and a desktop app for companies; schools and businesses already pay for night-shift transport.

EBVG-Hoot github.com/MarcusPicazo/ebvg-hoot
Real-time multiplayer quiz platform for classroom use. React and Firestore, structured with custom hooks separating game logic from presentation.

PROFESSIONAL EXPERIENCE

Lead Product Engineer · Escuela Berta Von Glümer Oct 2023 – Present

- Architected and shipped a suite of five React and TypeScript applications serving 200+ students and staff, defining the frontend architecture, shared component system and deployment pipeline end to end.
- Led delivery of EBVG-Hoot, a real-time multiplayer assessment platform, designing an event-driven data layer on Firestore onSnapshot to keep game state synchronised across concurrent players.
- Owned the 3D engineering track: three interactive learning environments in Three.js and WebGL — a solar system explorer, a colour-mixing art studio and a parkour game with custom collision physics — profiling the render loop and memory use to hold a stable frame rate on constrained school hardware.
- Established CIT-Sync, an internal platform for spatial management, hardware inventory and software requests, replacing manual spreadsheet workflows across departments.
- Gathered requirements directly from teachers and directors and translated them into working tools; built Duolingo-style English diagnostic tests as web apps.

Systems Engineer · Getting Out 2022 – 2023

- Built Python automation services, REST APIs and microservices that streamlined internal data processing workflows.
- Designed SQL data models and service-oriented architecture to support operational reporting.

Audio Engineer (Remote) · Black Fox Productions 2019 – 2025

- Delivered broadcast-ready audio post-production for international cinema chains, running asynchronous review cycles and signal routing across Pro Tools, Ableton Live, iZotope RX and Dante.

EARLIER EXPERIENCE

IT Systems Engineer · Cedros College 2021 – 2022

Network security and infrastructure administration across the corporate network — Active Directory, Fortinet, Windows Server, Group Policy.

Systems Engineer · CONACULTA / Secretaría de Cultura 2016 – 2017

Technical architecture support, QA testing and debugging of custom government software — Linux, SQL, Bash.

EDUCATION

Completed B.Eng. Computer Systems, Universidad Europea de Madrid · B.Sc. Music & Innovation, Universidad Panamericana
Coursework B.Sc. Physics, Universidad Autónoma Metropolitana (through final year) · M.Sc. Astrophysics

CERTIFICATIONS & LANGUAGES

Certifications IBM Full Stack Developer · IBM Machine Learning & Python · Google Digital Marketing · EF SET C2 English
Languages English (native) · Spanish (native) · Italian (B2) · Catalan (B2)